INCH-POUND
MIL-DTL-16377/52C(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/52B(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; SOLID-STATE, LIGHT, EXTENSION, 120 VOLTS, EXPLOSIONPROOF AND WATERTIGHT, SYMBOL 286SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 2 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

Distribution Statement A. Approved for public release: distribution is unlimited.

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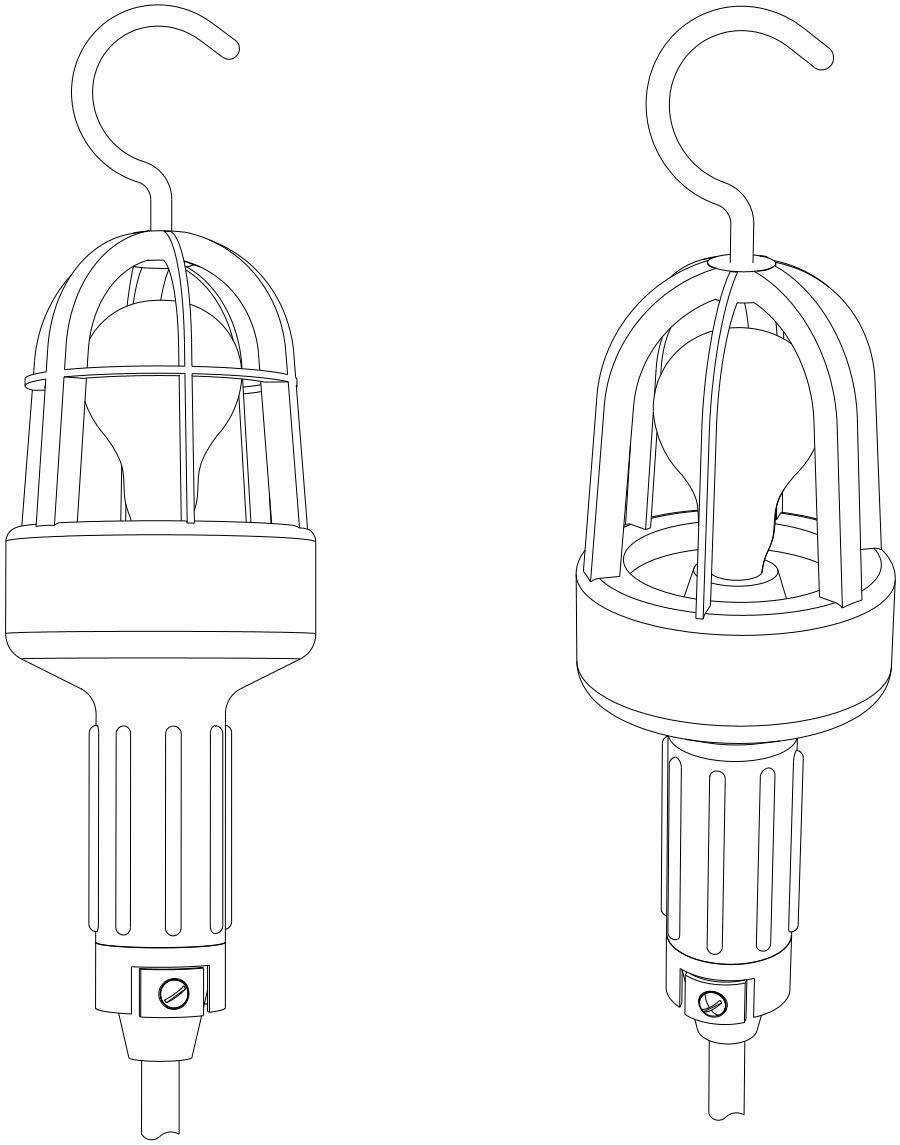


FIGURE 1. Typical view of symbol 286SSL.

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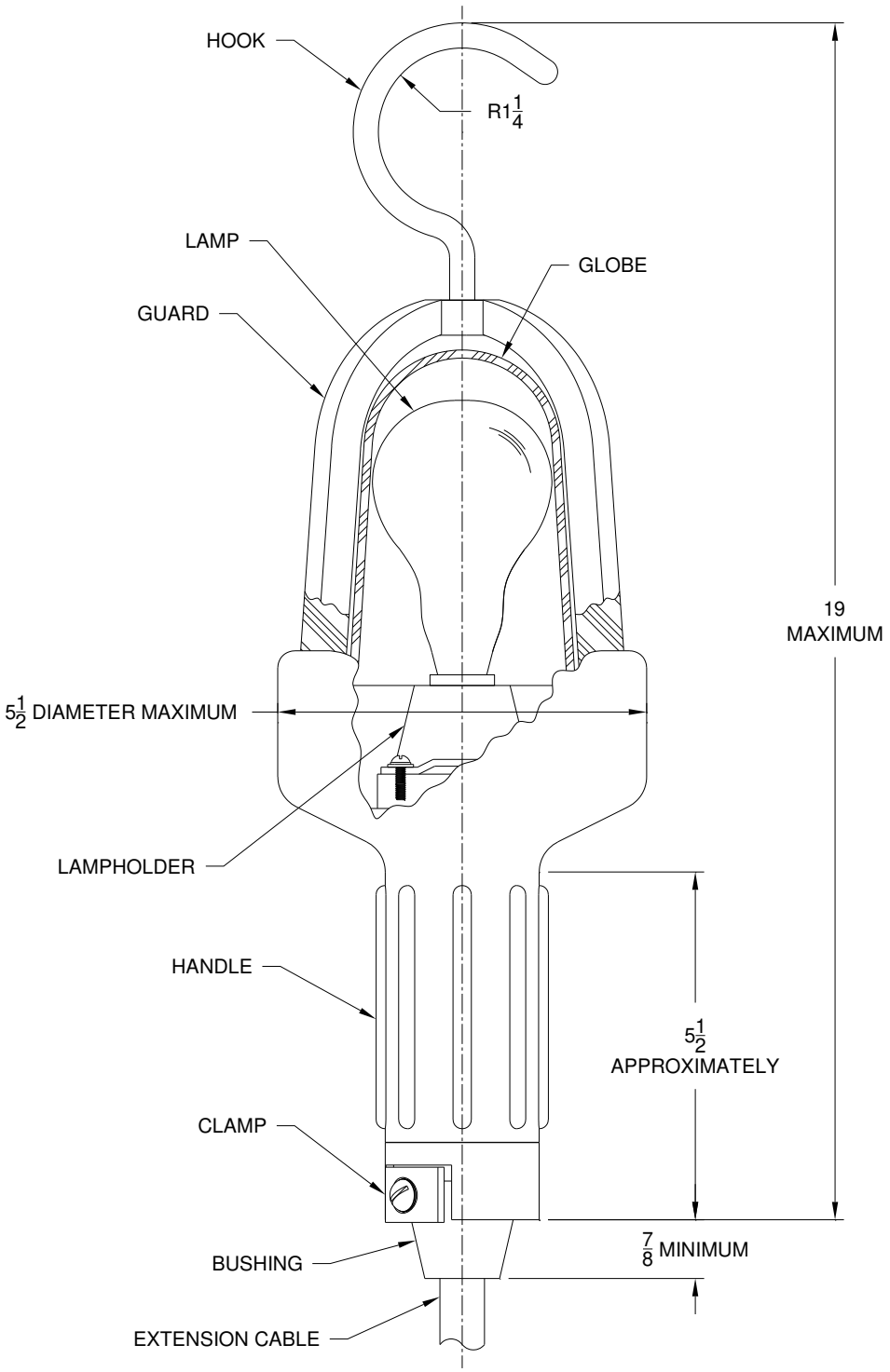


FIGURE 2. Typical view of symbol 286SSL.

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REQUIREMENTS

CLASSIFICATION

Type: III
Class: 1

SYMBOL NUMBER

286SSL: Light, extension, solid-state lighting (SSL), 120 volts, explosionproof and watertight.

DESIGN

Electrical rating:	50 watts maximum, 120 volts.
Dimensions and configuration:	See figures 1 and 2.
Metals:	All exposed metal parts shall be made of nonsparking metal, or the extension light shall be double insulated in accordance with Underwriters Laboratories (UL) standards.
Guard:	The guard shall be cast or fabricated of nonsparking metal with at least four ribs or shall be double insulated in accordance with UL standards.
Hook:	The design of the hook shall be such that it is permitted to turn 360 degrees and does not extend outside of the projected cylindrical surface of the light.
Handle:	The cable entrance end of the handle shall provide a watertight seal and shall relieve the strain on the cable and electrical conductors.
Light output:	1,200 lumens.
Enclosure:	The light shall be designed to be explosionproof and watertight and shall be so constructed that no parts will work loose in service. The enclosure shall be capable of withstanding the internal hydrostatic pressures resulting from explosions, without bursting or loosening at the joints. The enclosure shall withstand without rupture or permanent distortion, an internal hydrostatic test pressure.
Cable:	The light shall be furnished with 50 feet (plus 12, minus 0 inches) of three-conductor cable, type SO-16/3 or ST-16/3, in accordance with UL 62. The third conductor shall be used for grounding. If the extension light is double insulated in accordance with UL standards, the ground conductor is not required, and a two-conductor cable shall be provided.
Ambient temperature:	The light shall be capable of operating in an ambient temperature of 205 °F for 96 hours without damage or loosening of parts.
Weight:	The overall weight of the extension light less cable shall not exceed 5½ pounds.
Identification and information plates:	Identification and information plates shall be metal if the guard is metal or plastic if the guard is double insulated and shall be in accordance with MIL-DTL-16377. The following plates shall be furnished: (a) Identification plate. The following information shall appear on an identification plate that shall be securely attached to the housing: "Light, extension, explosionproof and watertight, shipboard use MIL-DTL-16377/52, Symbol 286SSL Manufacturer's name or trademark and model number of the item"

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(b) Warning plate. A warning plate bearing the following legend, in uppercase letters, shall be provided:

"WARNING

FOR USE IN THE FOLLOWING EXPLOSIVE ATMOSPHERES ONLY: GASOLINE, PETROLEUM, NAPHTHA, ALCOHOL, ACETONE, LACQUER, SOLVENT VAPORS, AND NATURAL GAS.

CAUTION

Disconnect from circuit before any maintenance that requires entry into the extension light. Keep all joints tight when in operation. If the connection to the electrical circuit is in an area where explosive gases are present, an explosionproof plug and matching receptacle shall be used. If the electrical connection is made outside of the hazardous area, a three-conductor grounded plug may be used. The light shall not be disassembled in a hazardous location."

Chromaticity: See 3.7.4 of MIL-DTL-16377.

Fail-safe circuit: See 3.6.17 of MIL-DTL-16377.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:.

Complete fixtures: See table I.

Conformance inspection:

Comparison inspection: Same as the first article inspection.

Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION

This light is intended to be used as a general-purpose extension light for detail illumination onboard ship and as a work light in areas where explosive gases may be present.

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TABLE I. First article and comparison inspection.

Tests	Requirement Paragraph of MIL-DTL-16377 or Reference to UL 844	Test Paragraph of MIL-DTL-16377 or Reference to UL 844	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Rough usage	UL 844	UL 844	Rough-usage test
Drop	UL 844	UL 844	---
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Ambient temperature	1/	1/	---
Grounding circuit	3.6.13.3	4.8.11	1/
Leakage current	3.6.13.5	4.8.13	---
Continuity of grounding	3.6.13.4	4.8.12	1/
Optical uniformity	3.7.7.1	4.8.4	---
Thermal shock	3.7.5.7	4.8.17.4	---
Breakage pattern	3.7.5.9	4.8.17.6	---
Color and luminous transmittance	3.7.3.7	4.8.6.3	---
Magnetic permeability	3.4.8	4.8.18	---
Cable pull	UL 844	UL 844	Strain relief test
Salt fog	3.5.15	4.8.10	---
Enclosure effectiveness	3.5.11	4.8.15	Explosionproof
Hydrostatic pressure	3.5.11	8.15.2	---
Photometrics	3.7.1	4.8.6	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Stress relief	3.8.5	4.8.5	---

NOTE:

1/ Only required for extension lights with exposed metal.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6230-2023-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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